TimeLens2 fixed training

Labels + reward · not inference

Fix	What it does
TimeLens2-93K	Multi-stage labels: propose → local dual-agent → consensus → semantic check → boundary refine
SFT → GRPO	Learn search syntax, then calibrate geometry
Reward	$R = R_{\mathrm{tIoU}} + R_{\mathrm{TW}} - \mathbf{1}_{\mathrm{invalid}}$

```
data:   multi-scale verification cascade
train:  set-valued interval geometry (match-free W₁)
```

Treats evidence as an interval set through supervision and optimization [Zhu et al., 2026, §3].

Gap: inference still one-shot uniform

```
(v, q)  →  uniform frames over full T   (e.g. 2 fps, cap ~512)
        →  single forward decode
        →  parse Ŷ
```

Train / data pipeline	Deployed inference
Propose → verify local → refine	**One** pass
Dense pixels on short clips	Thin sample on **all** of T
Multi-span built carefully	Easy to **skip** brief / late / repeated hits

```
irony:  labels made by hierarchical search
        model used without hierarchical search
```

Sparse evidence fails under uniform downsample [Liu et al., 2026]. Adaptive test-time allocation not first-class in TimeLens2 [Zhu et al., 2026].

Our idea · hierarchical test-time search

Training-free · frozen TimeLens2-4B · matched budget

```
Stage 1  RECALL     window video · embed-rank top-K
Stage 2  PRECISION  dense frames only on survivors · local ground
Stage 3  MERGE      global timestamps · gap-merge · set Ŷ
```

Not new SFT/GRPO. **Reallocate** frames toward sparse multi-span evidence.

Eval (primary)	
Bench	**OMTG** · 320 queries · 2–20 GT spans / query
Metric	**EtF1** (set-aware; cardinality matters)
Controls	one-shot · full multipass · uniform-window · **ours**
Rule	win only if better at **matched** compute (≤5% GPU time)

Router: Qwen3-VL-Embedding-2B. Grounder frozen. Optional later: QVHighlights regression.